

Ge 158 - Physical Principles of Remote Sensing Fall 2026

Tuesdays and Thursdays 1030-11:55am

Location: SMD 162

Instructors: Brent Minchew
Teaching Assistant: Leonid Pereiaslov

Description

This course is an introduction to the physics of remote sensing, with applications to the study of Earth, the Moon, and other planets and solar system bodies as well as to emerging industrial applications. We will cover the physical principles of optical, thermal, radar, and lidar remote sensing with an emphasis on the fundamental properties of electromagnetic waves; the principles of electromagnetic scattering from real and idealized materials, including various types of surfaces and vegetation; the interaction of electromagnetic radiation with the atmosphere; and thermal and microwave emission from various media. We will also discuss remote sensing platforms along with the essentials of orbital mechanics and data processing methods and tools.

Rationale for the course

Remote sensing is an indispensable tool in all disciplines of earth science and in many engineering disciplines. Science applications include observations of the structure, composition, and motion of solid surfaces, oceans, atmospheres, and cryospheres of the Earth, Moon, planets, and other solar system bodies. Governmental and commercial applications include land survey, surveillance, monitoring, hazard mitigation, and disaster response. There are also emerging commercial applications in navigation, infrastructure inspections, and industrial operations optimization. Companies from small (e.g., Capella Space) to Google are heavily invested in bolstering their remote sensing capabilities. In an age when data is the most valuable commodity, remote sensing has massive potential. This course will introduce the physical principles of the most commonly used remote sensing methods: optical, thermal, radar, and lidar. In addition, students will learn how to access, process, and analyze remote sensing data.

Prerequisites/ Corequisites

Prerequisite: None required, but an introduction to electromagnetism will be useful

Corequisite: None

Grading scheme

Grades will be based on three problem sets (psets) and a project. The psets are meant to be challenging but solvable within 5 hours by the average student not using AI.

The psets are expected to be submitted by each student individually. The onus to learn is on you: it is possible to solve the homeworks in minutes with AI tools, but you will not learn anything that way. Presumably, you came to Caltech to learn and not to outsource your thinking.

The class project will consist of a short abstract outlining the idea and a final report and presentation due the last week of class. A detailed project description is provided separately. Project expectations will differ for undergraduate and graduate students.

Grade breakdown:

- 3 psets x 10 % each = 30 % of final grade
- Project abstract = 25 % of final grade
- Final project report and presentation = 45 % of final grade
- **No exams or quizzes**

Collaborations and academic conduct

Collaboration is essential to science and engineering. We learn best when we feel free to share ideas and discuss problems. We encourage everyone to work together in groups for the final project and to submit one final report per group. However, we expect you all to contribute to the collaborations that you choose to be a part of, and to avoid taking unfair advantage of others.

We expect you to submit each assignment independently, written in your own words. You must state how you used AI tools to complete or check the assignments. You will only hurt yourself if you outsource everything to AI because you will not learn.

You will have to defend your final project in a live question and answer session. This is where you will have to demonstrate what you have learned.

Textbooks

- *Required:* None; all necessary materials will be provided
- *Recommended:* Rees, W. G., *Physical Principles of Remote Sensing* (provides some additional materials and follows roughly the same structure as the course). This [available in digital form](#) through the MIT Libraries.
- *Additional resources* (for those interested in learning more or delving into a particular application):
 - Richards, J. A., *Remote Sensing with Imaging Radar*
 - Ulaby, F. T., *Microwave Radar and Radiometric Remote Sensing*
 - F. T. Ulaby, R. K. Moor, and A. K. Fung, *Microwave Remote Sensing: Active and Passive*
 - van Zyl, J. and Kim, Y., *Synthetic Aperture Radar Polarimetry*
 - Griffiths, D. J., *Introduction to Electrodynamics*

Important dates from Caltech registrar for [Fall 2026](#):

- Sep 29, first day of class
- Oct 16, last day to add classes
- Nov 18, last day to drop classes
- Nov 26-27, Thanksgiving holiday
- Dec 1, last day of class
- Dec 7-11, final exam period (Good luck! There is no final in this course.)

Lecture Schedule (subject to modification)

- 0 - Syllabus and course overview, introduce remote sensing
- 1 - Review basic EM concepts and dielectric material properties
- 2 - Review Maxwell's Equations and EM waves

- 3 - Brief intro to polarization ([self-study; no in-class lecture](#))
- 4 - Reflection and transmission at planar interfaces: 1. Normal incidence
- 5 - Reflection and transmission at planar interfaces: 2. Oblique incidence
- 6 - Reflection and transmission in multi-layered materials ([self-study; no in-class lecture](#))
- 7 - Scattering from nonplanar surfaces
- 8 - Scattering from surfaces with random small-scale roughness (specifically, the ocean surface)
- 9 - Volume scattering
- 10 - Interaction of EM radiation w/ atmospheres, part 1
- 11 - Interaction of EM radiation w/ atmospheres, part 2
- 12 - Atmospheric scattering and black-body radiation
- 13 - Radiometry and radiative transfer
- 14 - Synthetic aperture radar (SAR)